Review Summary

Concepts & Comprehension

- Application of calculus in physics
  - Finding initial condition
- Simple harmonic motion (SHM), $x = A\cos(\omega t + \varphi)$, $E \propto A^2$
  - Angular frequency: $\omega = \sqrt{k/m} = 2\pi f$, $T = 2\pi/\omega$
- Fixed-axis rotation of rigid body, moment of inertia
  - Right-hand rule for judging direction
- Degree of freedom, Hamiltonian canonical equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \;\&\; \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Concepts & Comprehension

- Flux of electromagnetic field
- Electrostatic equilibrium of conductor: $E = \sigma/\varepsilon_0$ (outer surface)
- Divergency & Curl

$$\nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}, \nabla \times \mathbf{J} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ J_x & J_y & J_z \end{vmatrix}$$

- Electromotive force: $\mathcal{E} = \frac{\mathrm{d}W_{\mathrm{ne}}}{\mathrm{d}q}$
- Electrostatic field and induced electric field

$$\mathbf{E} = \mathbf{E}_{\mathrm{s}} + \mathbf{E}_{\mathrm{i}}, \nabla \times \mathbf{E}_{\mathrm{s}} = \mathbf{0}, \nabla \cdot \mathbf{E}_{\mathrm{i}} = 0$$

Equations & Formulas

	Linear	Circular/Rotational
Kinematics	$\mathbf{r}, \Delta\mathbf{r}$ $\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$ $\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$	$\boldsymbol{\theta}, \Delta\boldsymbol{\theta}$ $\boldsymbol{\omega} = \frac{\mathrm{d}\boldsymbol{\theta}}{\mathrm{d}t}$ $\boldsymbol{\alpha} = \frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t} = \frac{\mathrm{d}^2\boldsymbol{\theta}}{\mathrm{d}t^2}$
Dynamics	$\mathbf{F} = m\mathbf{a} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}$ $\mathbf{p} = m\mathbf{v}, K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$ $W = \Delta K = \int \mathbf{F} \cdot \mathrm{d}\mathbf{r}$	$\boldsymbol{\tau} = I\boldsymbol{\alpha} = \frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t}$ $\mathbf{L} = I\boldsymbol{\omega}, K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$ $W = \Delta K = \int \boldsymbol{\tau} \cdot \mathrm{d}\boldsymbol{\theta}$

$\times \mathbf{r}$

m

I

Equations & Formulas

- Potential & conservative (force/field), gradient operator

$$\mathbf{F} = -\nabla U, \nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}$$

- Equilibrium & stability in 1-dimensional case

$$F = -\frac{\mathrm{d}U}{\mathrm{d}x} = 0, \frac{\mathrm{d}^2 U}{\mathrm{d}x^2} > 0, < 0 \text{ or } = 0$$

- Potential energy: $U = -\frac{GMm}{r}$ (gravitation); $U = \frac{1}{2}kx^2$ (SHM)
- Lagrangian/Hamiltonian function, general momentum

$$L = K - U, p_i = \frac{\partial L}{\partial \dot{q}_i}, H = K + U$$

Equations & Formulas

- Maxwell's equations

$$\oint \mathbf{D} \cdot \mathrm{d}\mathbf{A} = q_{\mathrm{in}}, \nabla \cdot \mathbf{D} = \rho$$

$$\oint \mathbf{B} \cdot \mathrm{d}\mathbf{A} = 0, \nabla \cdot \mathbf{B} = 0$$

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\int \frac{\partial \mathbf{B}}{\partial t} \cdot \mathrm{d}\mathbf{A}, \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\oint \mathbf{H} \cdot \mathrm{d}\mathbf{l} = I_{\mathrm{in}} + \int \frac{\partial \mathbf{D}}{\partial t} \cdot \mathrm{d}\mathbf{A}, \nabla \times \mathbf{H} = \mathbf{j} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\mathbf{D} = \varepsilon \mathbf{E}, \mathbf{H} = \frac{\mathbf{B}}{\mu}, I = \int \mathbf{j} \cdot \mathrm{d}\mathbf{A}$$

- Electromagnetic potential

$$\mathbf{E}_{\mathrm{s}} = -\nabla V$$

$$\mathbf{B} = \nabla \times \mathbf{A}_{\mathrm{M}}$$

$$\mathbf{E}_{\mathrm{i}} = -\frac{\partial \mathbf{A}_{\mathrm{M}}}{\partial t}$$

- Electromagnetic force
  - For charge: $\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$
  - For current: $\mathrm{d}\mathbf{F} = I\mathrm{d}\mathbf{l} \times \mathbf{B}$
- Electromagnetic energy

$$u_{\mathrm{E}} = \frac{1}{2}\varepsilon E^2, u_{\mathrm{B}} = \frac{B^2}{2\mu}$$

Methodology of calculation

- Establishing and solving a simple ODE

$$F = ma = m\frac{\mathrm{d}v}{\mathrm{d}t} = F(v) \;\rightarrow\; \frac{1}{F(v)}\mathrm{d}v = m\mathrm{d}t \;\rightarrow\; v = v(t)$$

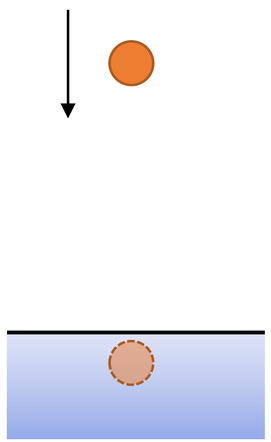
- Calculation of moment of inertia: cylindrical integral

$$I = \int r^2 \mathrm{d}m\,, \mathrm{d}m = \rho \mathrm{d}V, \mathrm{d}V = 2\pi r h \mathrm{d}r$$

- Application of Gauss's law and Ampère's law
  - Calculation of capacity: $C = Q/V, V = \int E(r)\,\mathrm{d}r$
- Kirchhoff's rules

$$\sum_i I_i = 0 \text{ (junctions)}; \sum V = \sum(\mathcal{E} - Ir) \text{ (loops)}$$

[Example] A small ball of mass m falls vertically, and there is a resistance (阻力) from the air, which is proportional to the speed: $f_1 = -b_1 v$. (a) Prove that it will approach a constant speed if the time is enough. (b) After that it falls into water, and the drag force of the water follows: $f_1 = -b_2 v^2$. If its gravity is cancelled by the buoyancy (浮力), determine its motional equation in water before the lowest position.



[Solution] (a) The dynamic of the ball in the air:

$$ma = mg - b_1 v = m\frac{\mathrm{d}v}{\mathrm{d}t}$$

Set $v' = v - mg/b_1$:

$$-b_1 v' = m\frac{\mathrm{d}v'}{\mathrm{d}t} \quad \rightarrow \quad -\frac{b_1}{m}\mathrm{d}t = \frac{\mathrm{d}v'}{v'}$$

$$-\frac{b_1}{m}t = \ln v' + C \quad \rightarrow \quad \frac{v - mg/b_1}{mg/b_1} = \mathrm{e}^{-\frac{b_1}{m}t}$$

$$v = \frac{mg}{b_1}\left(1 - \mathrm{e}^{-\frac{b_1}{m}t}\right)$$

So the speed approaches mg/b_1.

(b) The dynamic of the ball in water:

$$ma = -b_2 v^2 = m\frac{\mathrm{d}v}{\mathrm{d}t}$$

$$-\frac{b_2}{m}\mathrm{d}t = \frac{\mathrm{d}v}{v^2}$$

$$-\frac{b_2}{m}t = \frac{b_1}{mg} - \frac{1}{v}$$

$$v = \frac{1}{b_2 t/m - b_1/mg}$$

If the water surface is $x = 0$:

$$x = \frac{m}{b_2}\ln\frac{b_1/g}{b_2 t - b_1/g}$$

[Example] The period of a simple pendulum on the ground is 2.5s. The escape speed (逃逸速度) on the ground is 1.12×10^4m/s. The Mars (火星) has 53% of the radius of the earth, and 11% of the mass of the earth. Determine the period of the pendulum and the escape speed on the surface of the Mars.

[Solution] From $T = 2\pi\sqrt{L/g}$:

$$\frac{T'}{T} = \left(\frac{g'}{g}\right)^{-\frac{1}{2}} = \left(\frac{GM'/r'^2}{GM/r^2}\right)^{-\frac{1}{2}}$$

$$= \left(\frac{M'}{M}\right)^{-\frac{1}{2}}\frac{r'}{r} = 1.6$$

$$T = 4.0\text{s}$$

According to the conservation of mechanical energy:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = 0$$

where v is the escape speed. Hence

$$v = \sqrt{\frac{2GM}{r}}$$

$$\frac{v'}{v} = \left(\frac{GM'/r'}{GM/r}\right)^{\frac{1}{2}}$$

$$= \left(\frac{M'}{M}\right)^{\frac{1}{2}}\left(\frac{r'}{r}\right)^{-\frac{1}{2}} = 0.46$$

$$v = 5.1\times10^3\text{m/s}$$

[Example] Show that the moment of inertia of a uniform annular cylinder to the axis through the center along the axis of symmetry is

$$I = M(R_1^2 + R_2^2)/2$$

with the inner radius R_1, outer radius R_2, and mass M.

$I = mr^2$

[Solution] Divide the annular cylinder into thin concentric cylindrical rings of thickness $\mathrm{d}r$. The mass of the ring of radius $r(R_1 < r < R_2)$ is

$$\mathrm{d}m = 2\pi r\rho h\mathrm{d}r$$

where ρ is the density and h is the height of the entity.
Every part of this ring has a distance of r away from the axis, so the moment of inertia of this ring is

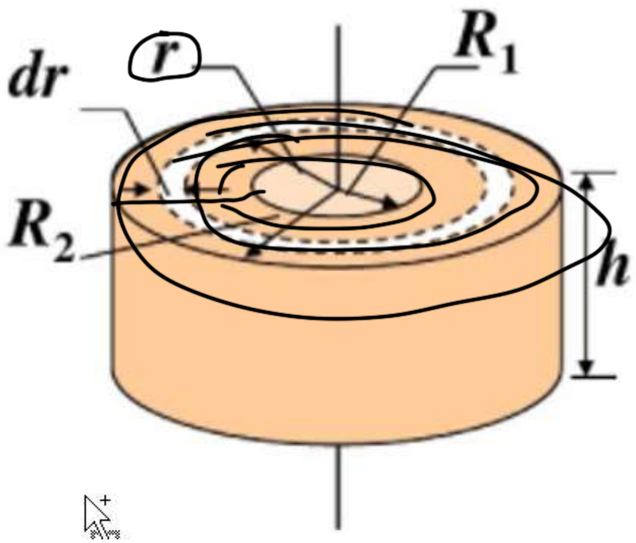
$$\mathrm{d}I = r^2\mathrm{d}m = 2\pi\rho h r^3\mathrm{d}r$$

Make integral of all the rings from R_1 to R_2:

$$I = 2\pi\rho h\int_{R_1}^{R_2} r^3\mathrm{d}r = \frac{\pi\rho h}{2}(R_2^4 - R_1^4)$$

Because $M = \pi\rho h(R_2^2 - R_1^2)$, substitute it into I and we obtain:

$$I = M(R_1^2 + R_2^2)/2$$



[Example: multi-layered dielectric] A cylindrical capacitor of length l, inner radius R_1 and outer radius R_2 is filled with two layers of dielectric of ε_1 inside and ε_2 outside. The radius of the interface between the dielectric layers is R. Find the capacitance of the capacitor.

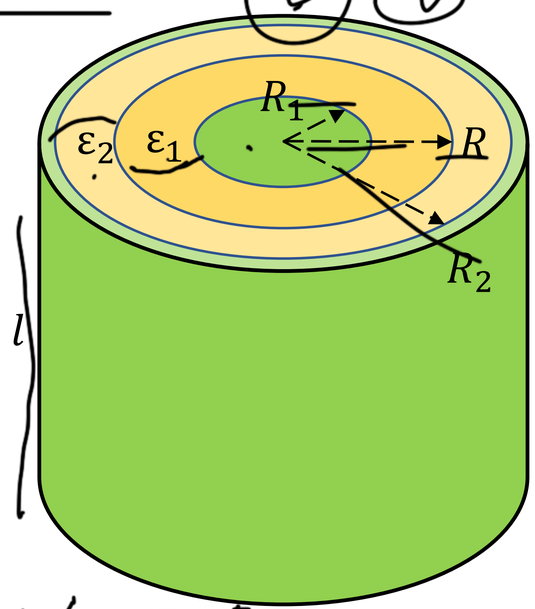
$C = \frac{Q}{V}$

[Solution] Assume the capacitor has a net charge of Q, By use of Gauss's law with dielectric we have

$\int E \cdot A = \frac{q_{in}}{\varepsilon_k}$

$$D = \frac{Q}{2\pi r l} \rightarrow E = \begin{cases} \frac{Q}{2\pi r \varepsilon_1 l} \ldots\ldots R_1 \le r < R \\ \frac{Q}{2\pi r \varepsilon_2 l} \ldots\ldots R \le r < R_2 \end{cases}$$

$D = E\varepsilon$



Hence the potential difference between R_1 and R_2 is

$$V = \int_{R_1}^{R} \frac{Q}{2\pi r \varepsilon_1 l} dr + \int_{R}^{R_2} \frac{Q}{2\pi r \varepsilon_2 l} dr = \frac{Q}{2\pi l}\left[\frac{\ln(R/R_1)}{\varepsilon_1} + \frac{\ln(R_2/R)}{\varepsilon_2}\right]$$

$V = E d$

$$C = \frac{Q}{V} = \frac{2\pi l \varepsilon_1 \varepsilon_2}{\varepsilon_2 \ln(R/R_1) + \varepsilon_1 \ln(R_2/R)}$$

$$\frac{1}{C} = \frac{\ln(R/R_1)}{2\pi \varepsilon_1 l} + \frac{\ln(R_2/R)}{2\pi \varepsilon_2 l} = \frac{1}{C_1} + \frac{1}{C_2}$$

It is equivalent to two capacitors connected in series.

[Example] A long straight cylindrical wire of radius R and permeability μ carries current of I. Determine the energy of magnetic field inside the wire per unit length.

$\int B \cdot dr = I_{in} \cdot \mu$

[Solution] Select a circular loop perpendicular to the axis with its center on the axis: r

$$\oint \mathbf{H} \cdot \mathrm{d}\mathbf{l} = 2\pi r H = I_{\text{in}} \rightarrow \quad B = \mu H = \frac{\mu}{2\pi r}\frac{\pi r^2}{\pi R^2} I$$

$H = \frac{B}{\mu}$

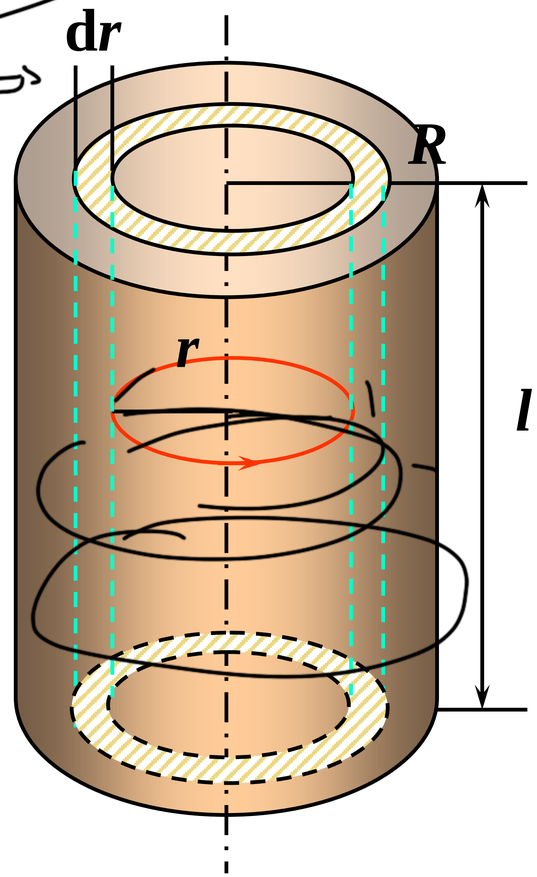
The magnetic energy density is

$$u = \frac{B^2}{2\mu} = \frac{\mu I^2 r^2}{8\pi^2 R^4}$$

Divide the wire into coaxial tubes of thickness $\mathrm{d}r$:

$$\mathrm{d}V = 2\pi r l \mathrm{d}r \quad \rightarrow \quad \mathrm{d}U = u\mathrm{d}V = \frac{\mu I^2 l}{4\pi R^4} r^3 \mathrm{d}r$$

$$U = \int_0^R \frac{\mu I^2 l}{4\pi R^4} r^3 \mathrm{d}r = \frac{\mu I^2}{16\pi} l \quad \rightarrow \quad \frac{U}{l} = \frac{\mu I^2}{16\pi}$$



[Discussion] If we try to calculate the energy of the whole space, we will find it approaching infinity. This is because we assume the wire is infinitely long, which is impossible.

多选题 1分

设置

Which of the following statements are true?

A The path (路程) and the magnitude of displacement approaches equal in an instantaneous process.

B If the acceleration of a particle is $a = \cos t$, the motional function is $x = -\cos t$. (Assume that the particle is initially rest at $x = 0$.)

C In an arbitrary (任意) time interval, there must exist a moment when the instantaneous speed is equal to the average speed of the whole interval.

D There are two motional functions $x_1 = 2\cos \pi t$, $x_2 = t - \ln t$. The ratio of their velocities v_1/v_2 approaches 2 at about $t = 1$.

提交

填空题 4分

设置

The potential function is

$$U = \frac{1}{4}x^4 - x^3 - \frac{1}{2}x^2 + 3x$$

The number of the stable equilibrium(ia) is [填空1].

The minimum positional value of the stable equilibrium(ia) is x=[填空2].

The number of the stable equilibrium(ia) is [填空3].

The minimum positional value of the stable equilibrium(ia) is x=[填空4].

$$\frac{\partial U}{\partial x} = 0$$

$$\frac{\partial^2 U}{\partial x^2} = 0 \quad \times$$

$$> 0 \quad \text{—}$$

$$< 0 \quad \text{—}$$

作答

填空题 3分

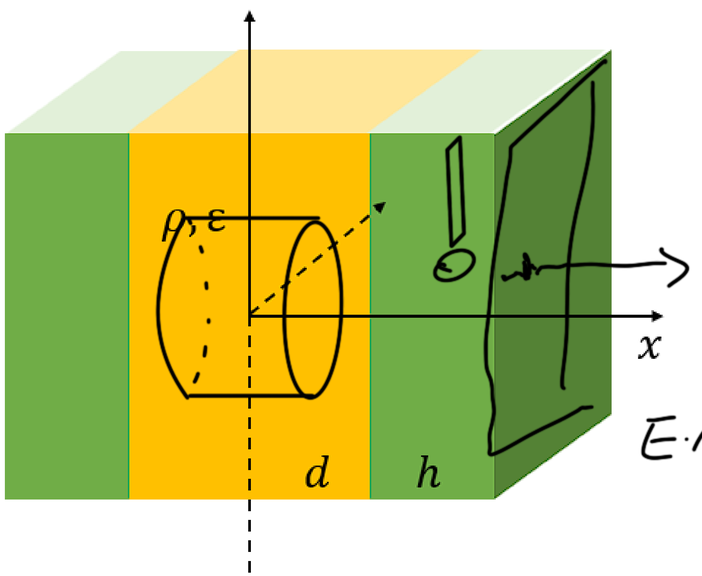
设置

An infinite insulator plate of permittivity ε and thickness $2d$ carries a uniform charge density ρ. Two other infinite conductor plates of thickness h, which carry no net charge, are parallel together with it.

The electric field for $0 < x < d$ is [填空1].

The electric field for $x > d + h$ is [填空2].

The surface charge density at $x = d + h$ is [填空3].



$\Phi = \int E \cdot dA = \frac{q_{in}}{\varepsilon}$

$E \cdot \cancel{4\pi r^2} = \frac{d\Phi}{dt} = \varepsilon$

$\int E \cdot dA = \frac{q_{in}}{\varepsilon}$

$2\pi r h$

(1) $E \cdot 2A = \frac{\rho A \cdot 2x}{\varepsilon}$, $\frac{\rho x}{\varepsilon}$

(2) $E \cdot 2A = \frac{\rho \cdot 2A \cdot d}{\varepsilon_0}$, $\frac{\rho d}{\varepsilon_0}$

(3) $E \cdot A = \frac{\sigma \cdot A}{\varepsilon_0}$, $\frac{\sigma}{\varepsilon_0} = \frac{\rho d}{\varepsilon_0}$, $\sigma = \rho d$

$E \cdot A = \frac{q_{in}}{\varepsilon}$

$E \cdot 2A = \frac{q_{in}}{\varepsilon}$

作答

填空题 3分

设置

The magnetic vector potential of a field is

$$\mathbf{A}_{\mathrm{M}} = \ln x\,\mathbf{j}$$

The magnetic field $\mathbf{B}$ is [填空1] $\mathbf{i}$ + [填空2] $\mathbf{j}$ + [填空3] $\mathbf{k}$.

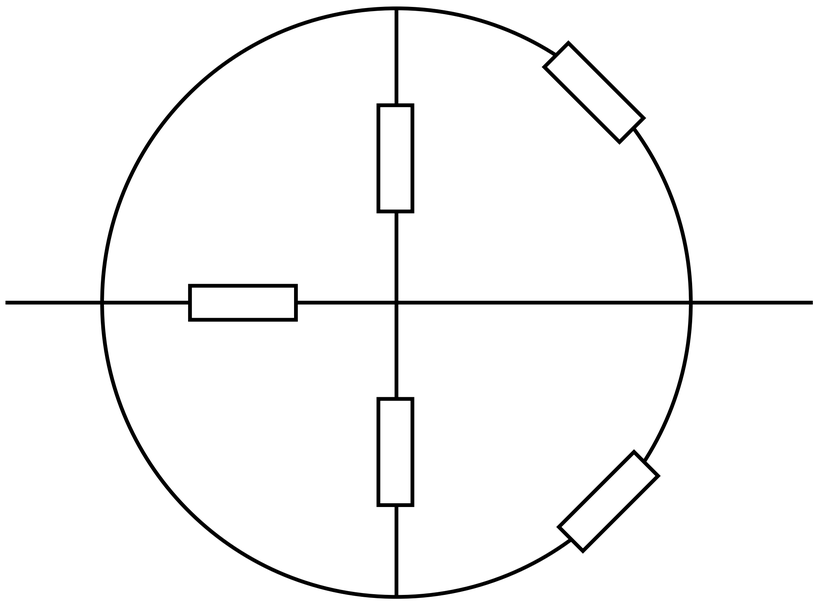
$B = \nabla \times A_m$

作答

填空题 1分

设置

As is shown, the resistance of each resistor is R. Determine the equivalent resistance between A and B: [填空1] R.



作答